

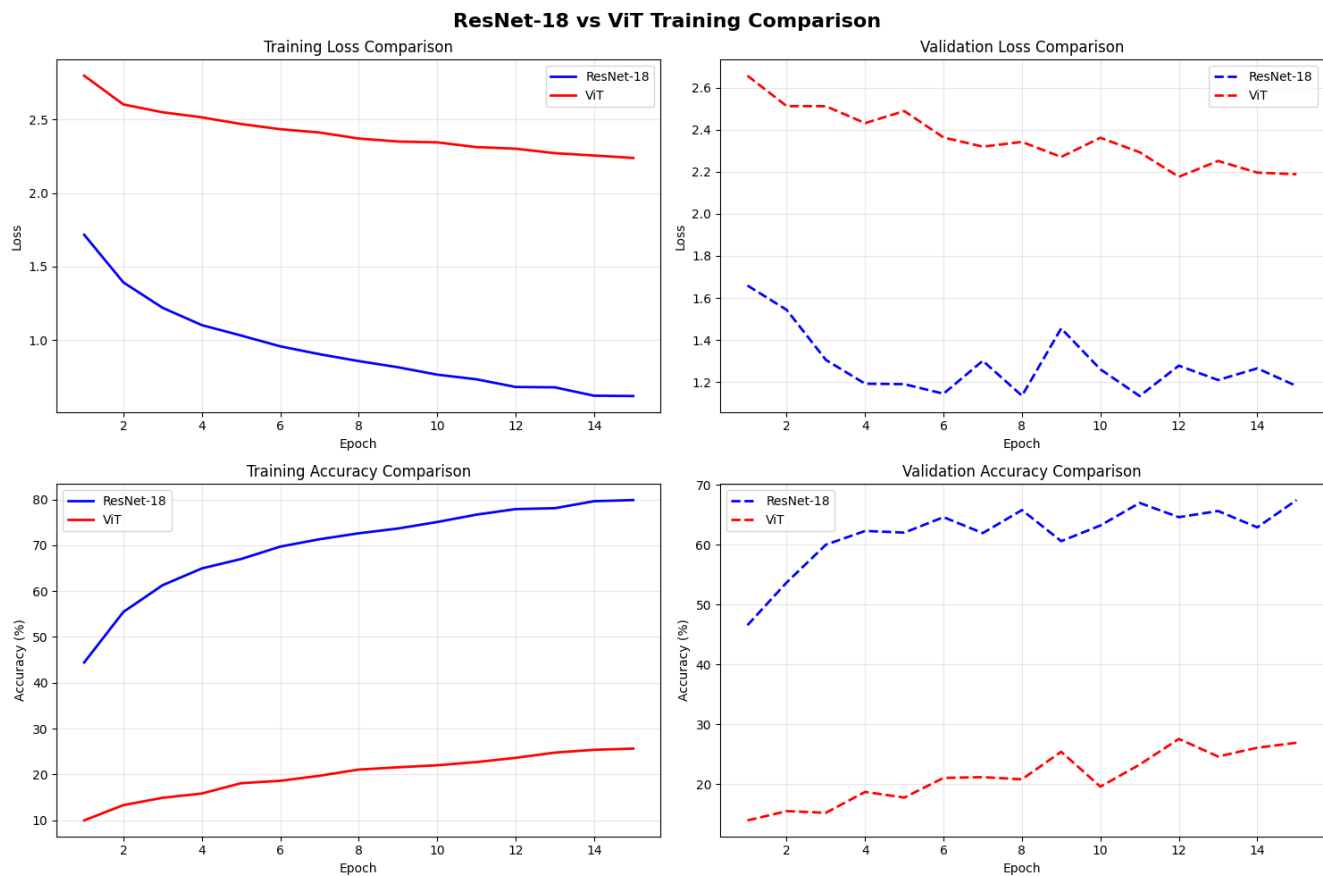
Assignment 01: ResNet-18 vs ViT

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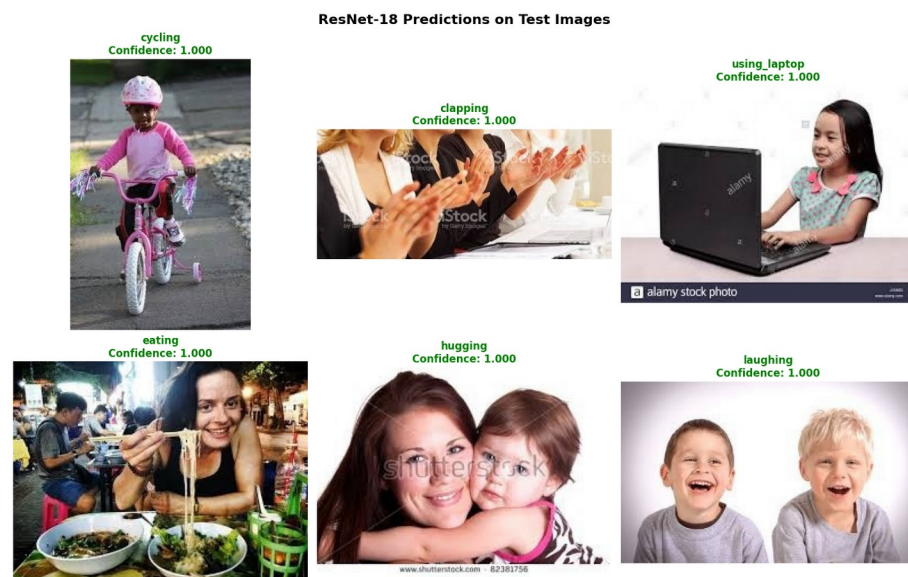
Due Date: 28Aug2025

In this project, we compare the effectiveness of ResNet-18 and ViT when classifying on the Human Action Recognition Dataset.

Here is the graph of the training progress between the 2 models:



Here are some predictions of both models on test images:



ViT Predictions on Test Images

laughing
Confidence: 0.855



cycling
Confidence: 0.846



sleeping
Confidence: 0.731



laughing
Confidence: 0.839



cycling
Confidence: 0.830



cycling
Confidence: 0.829



Conclusion:

The training of ResNet-18 was about 12x faster than ViT. The inference speed of ResNet-18 was on average 5.88ms/batch of 32 images versus 71.73ms per batch of 32 images for ViT. Also the accuracy was higher during training for ResNet-18 vs ViT at 79.88% for ResNet-18 vs 25.62% for ViT. So at least in the way the models were set up, ResNet-18 is much faster and performs at a much greater accuracy than ViT. This might be due in part to the way the ViT model was set up. Possibly using different pretrained weights or altering the model structure would improve accuracy on the HAR dataset.